

Problem 1

Let $S(n, k)$ denote the number of ways to partition $\{1, 2, \dots, n\}$ into k nonempty, unlabeled subsets. Give a story proof for each of the following identities.

(a)

prove

$$S(n+1, k) = S(n, k-1) + kS(n, k).$$

Consider partitioning the set

$$\{1, 2, \dots, n, n+1\}$$

into k nonempty, unlabeled subsets.

Case 1: The element $n+1$ forms a subset by itself.

Then the remaining n elements

$$\{1, 2, \dots, n\}$$

must be partitioned into $k-1$ nonempty subsets. There are

$$S(n, k-1)$$

ways.

Case 2: The element $n+1$ does not form a subset by itself.

First partition

$$\{1, 2, \dots, n\}$$

into k nonempty subsets. This can be done in

$$S(n, k)$$

ways.

Then $n+1$ can be placed into any one of these k subsets, giving

$$kS(n, k)$$

possibilities.

the two cases are disjoint and cover all possible partitions

$$\implies S(n+1, k) = S(n, k-1) + kS(n, k)$$

(b)

prove

$$S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k-1).$$

Consider partitioning

$$\{1, 2, \dots, n\}$$

into k nonempty, unlabeled subsets.

Focus on the subset containing the element n .

Let j denote the number of elements that are not in the same subset as n . These j elements must form the other $k - 1$ subsets.

First, choose these j elements from

$$\{1, 2, \dots, n - 1\}.$$

There are

$$\binom{n-1}{j}$$

ways to choose them. $\therefore$ partition these j chosen elements into $k - 1$ nonempty subsets.

$$\therefore S(j, k - 1)$$

ways.

The remaining $n - 1 - j$ elements must all belong to the same subset as n , so there are no further choices.

Therefore, for a fixed value of j , the number of partitions is

$$\binom{n-1}{j} S(j, k - 1).$$

the j elements must form $k - 1$ nonempty subsets

$$\implies j \geq k - 1.$$

$$j \leq n - 1.$$

$$\implies S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k - 1)$$

Problem 2

A nonrepeat password is a nonempty sequence formed from m distinct symbols, where no symbol may appear more than once. The length of the password is ℓ , which could be any integer from 1 to m . Suppose that one password is chosen uniformly at random from all nonrepeat passwords. Let p_m be the probability that the selected password uses all m symbols.

(a)

For a password of length ℓ , the number of the ℓ -length password is

$$\binom{m}{\ell} * \ell! = \frac{m!}{(m-\ell)!}$$

since the total number of nonrepeat passwords is

$$\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}.$$

The number of m -length password is

$$m!$$

$$\therefore p_m = \frac{m!}{\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}}$$

(b)

$$p_m = \frac{m!}{\sum_{\ell=1}^m \frac{m!}{(m-\ell)!}}.$$

$$\Rightarrow p_m = \frac{m!}{m! \sum_{\ell=1}^m \frac{1}{(m-\ell)!}}.$$

$$\Rightarrow p_m = \frac{1}{\sum_{\ell=1}^m \frac{1}{(m-\ell)!}}.$$

Let

$$j = m - \ell.$$

Thus

$$j \in [0, m-1]$$

$$\sum_{\ell=1}^m \frac{1}{(m-\ell)!} = \sum_{j=0}^{m-1} \frac{1}{j!}.$$

$$\Rightarrow p_m = \frac{1}{\sum_{j=0}^{m-1} \frac{1}{j!}}$$

From Taylor series expansion :

$$e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}.$$

When $x = 1$:

$$e = \sum_{j=0}^{\infty} \frac{1}{j!}.$$

$$\Rightarrow \lim_{m \rightarrow \infty} \sum_{j=0}^{m-1} \frac{1}{j!} = e.$$

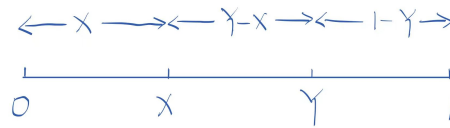
Thus

$$\boxed{\lim_{m \rightarrow \infty} p_m = \frac{1}{e}}.$$

Problem 3

(Geometry Probability) You get a stick and break it randomly into three pieces. Two breaking points are chosen independently and uniformly. What is the probability that you can make a triangle using such three pieces? assume there are two breaking point, the coordinates are X and Y ,

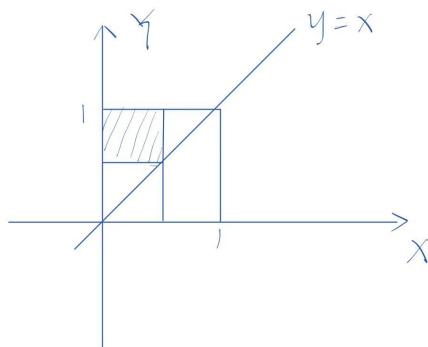
$$0 < X < Y < 1.$$



$$\begin{cases} X + (Y - X) \geq (1 - Y) \\ X + (1 - Y) \geq (Y - X) \\ (Y - X) + (1 - Y) \geq X \end{cases}$$

$$\Rightarrow \begin{cases} X \leq \frac{1}{2} \\ Y \geq \frac{1}{2} \\ Y \geq X \end{cases}$$

the total area is the square, the region satisfying the requirements is the shaded area.



$$\therefore A_{\text{total}} = 1 * 1 = 1.$$

$$A_{\text{shaded}} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

$$\Rightarrow P = \frac{A_{\text{shaded}}}{A_{\text{total}}} = \frac{\frac{1}{4}}{1} = \frac{1}{4}.$$

Problem 4

Let

$$\mathbf{p} = (p_1, p_2, \dots, p_{365}), \quad \sum_{j=1}^{365} p_j = 1.$$

Recall

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < \dots < j_k \leq n} x_{j_1} \cdots x_{j_k}.$$

(a)

Assume the event that at least one birthday match is A

The probability that all k birthdays are distinct is B

$$P(\mathbf{B} \mid \mathbf{p}) = k! \sum_{1 \leq j_1 < \dots < j_k \leq 365} p_{j_1} \cdots p_{j_k}.$$

$$\therefore P(\text{all distinct} \mid \mathbf{p}) = k! e_k(p_1, \dots, p_{365}).$$

$$\therefore P(\text{at least one match} \mid \mathbf{p}) = 1 - k! e_k(p_1, \dots, p_{365})$$

(b)

For $k = 2$,

$$P(\text{match}) = \sum_{j=1}^{365} p_j^2.$$

If

$$p_j = \frac{1}{365},$$

then

$$P(\text{match}) = 365 \left(\frac{1}{365} \right)^2 = \frac{1}{365}.$$

In an extreme case,

$$p_1 = 1, \quad p_2 = \dots = p_{365} = 0,$$

so for $k \geq 2$,

$$P(\text{at least one match}) = 1.$$

Thus, concentrating probability on fewer days increases collisions, while

$$p_1 = p_2 = \dots = p_{365} = \frac{1}{365}$$

spreads birthdays most evenly and minimizes the probability of a match.

(c)

First,

$$e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n).$$

Define

$$r_1 = r_2 = \frac{p_1 + p_2}{2}, \quad r_j = p_j, \quad 3 \leq j \leq 365.$$

Then

$$r_1 + r_2 = p_1 + p_2.$$

Also, by AM-GM,

$$\frac{p_1 + p_2}{2} \geq \sqrt{p_1 p_2},$$

so

$$r_1 r_2 = \left(\frac{p_1 + p_2}{2} \right)^2 \geq p_1 p_2.$$

Hence

$$\begin{aligned} e_k(\mathbf{r}) - e_k(\mathbf{p}) &= (r_1 r_2 - p_1 p_2) e_{k-2}(p_3, \dots, p_{365}) \\ &= \left[\left(\frac{p_1 + p_2}{2} \right)^2 - p_1 p_2 \right] e_{k-2}(p_3, \dots, p_{365}) \\ &= \frac{(p_1 - p_2)^2}{4} e_{k-2}(p_3, \dots, p_{365}) \\ &\geq 0. \end{aligned}$$

Therefore,

$$e_k(\mathbf{r}) \geq e_k(\mathbf{p}).$$

Using part (a),

$$P(\text{match} \mid \mathbf{p}) = 1 - k! e_k(\mathbf{p}),$$

and

$$P(\text{match} \mid \mathbf{r}) = 1 - k! e_k(\mathbf{r}).$$

Thus,

$$P(\text{match} \mid \mathbf{p}) \geq P(\text{match} \mid \mathbf{r})$$

If $p_1 \neq p_2$, then

$$(p_1 - p_2)^2 > 0,$$

and hence, when $e_{k-2} > 0$,

$$P(\text{match} \mid \mathbf{p}) > P(\text{match} \mid \mathbf{r}).$$

Thus averaging two unequal probabilities decreases the match probability. Repeated averaging gives

$$p_1 = p_2 = \dots = p_{365}.$$

Since

$$\sum_{j=1}^{365} p_j = 1,$$

we obtain

$$365p_j = 1,$$

so

$$p_j = \frac{1}{365}, \quad j = 1, \dots, 365$$

Therefore,

$$P(\text{at least one birthday match}) \text{ is minimized at } p_j = \frac{1}{365}$$

Problem 5

A card collection contains five different types of cards. Each independent draw produces card type i with probability p_i , where

$$(p_1, p_2, p_3, p_4, p_5) = (0.05, 0.15, 0.20, 0.25, 0.35).$$

Let Q_n denote the probability of collecting all five card types after n independent draws.

For each $i \in \{1, \dots, 5\}$, define

$$E_i = \{\text{card type } i \text{ does not appear in the first } n \text{ draws}\}.$$

For any $I \subseteq \{1, \dots, 5\}$, define

$$E_I = \bigcap_{i \in I} E_i.$$

(a)

Collecting all five card types means that none of the events $E_1, \dots, E_5$ occurs. Hence

$$Q_n = 1 - P\left(\bigcup_{i=1}^5 E_i\right).$$

By the inclusion-exclusion principle,

$$Q_n = 1 - \sum_i P(E_i) + \sum_{i < j} P(E_i \cap E_j) - \sum_{i < j < k} P(E_i \cap E_j \cap E_k) + \dots.$$

Based on the question stem

$$E_I = \bigcap_{i \in I} E_i$$

For one draw,

$$P(\text{no type in } I) = 1 - \sum_{i \in I} p_i.$$

the n draws are independent,

$$P(E_I) = \left(1 - \sum_{i \in I} p_i\right)^n.$$

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} P(E_I).$$

$$\Rightarrow Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n$$

For $I = \emptyset$,

$$E_{\emptyset} = \Omega, \quad P(E_{\emptyset}) = 1,$$

so the term corresponding to $I = \emptyset$ gives the initial 1 in the inclusion-exclusion formula.

(b)

Using

$$(p_1, p_2, p_3, p_4, p_5) = (0.05, 0.15, 0.20, 0.25, 0.35),$$

we compute

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n.$$

A numerical search gives

$$Q_{45} = 0.8998932711 < 0.90,$$

while

$$Q_{46} = 0.9049652386 > 0.90.$$

Hence the smallest integer satisfying

$$Q_n \geq 0.90$$

is

$$n = 46$$

The code and terminal output are shown below.

```

1  import itertools
2
3  p = [0.05, 0.15, 0.20, 0.25, 0.35]
4
5  def Q(n):
6      total = 0.0
7
8      for r in range(6):
9          for I in itertools.combinations(range(5), r):
10
11              s = sum(p[i] for i in I)
12
13              total += (-1)**r * (1 - s)**n
14
15  return total
16
17  n = 1
18
19  while Q(n) < 0.90:
20      n += 1
21      print("smallest n =", n)
22      print("Q_{ } = {:.10f}".format(n - 1, Q(n - 1)))
23      print("Q_{ } = {:.10f}".format(n, Q(n)))

```

```

jasmin@localhost:~/latex$ ~/.venvs/probability/bin/python -u "/home/jasmin/latex/SI140/code1.py"
Q_46 = 0.9049652386
jasmin@localhost:~/latex$ ~/.venvs/probability/bin/python -u "/home/jasmin/latex/SI140/code1.py"
smallest n = 46
Q_45 = 0.8998992711
Q_46 = 0.9049652386
jasmin@localhost:~/latex$

```

Figure 1: Code and terminal output for part (b).

(c)

$$p_1 = p_2 = p_3 = p_4 = p_5 = \frac{1}{5}.$$

If $|I| = k$

$$\implies \sum_{i \in I} p_i = \frac{k}{5}.$$

$$\begin{aligned} \implies Q_n &= \sum_{k=0}^5 (-1)^k \binom{5}{k} \left(1 - \frac{k}{5}\right)^n. \\ &= 1 - 5 \left(\frac{4}{5}\right)^n + 10 \left(\frac{3}{5}\right)^n - 10 \left(\frac{2}{5}\right)^n + 5 \left(\frac{1}{5}\right)^n. \end{aligned}$$

notice that

$$Q_{17} = 0.8891009573 < 0.90,$$

$$Q_{18} = 0.9109429198 > 0.90.$$

the smallest integer in the uniform case is

$$n = 18$$

For comparison,

$$n_{\text{uniform}} = 18, \quad n_{\text{unequal}} = 46.$$

The probabilities Q_n for the two cases are plotted below.

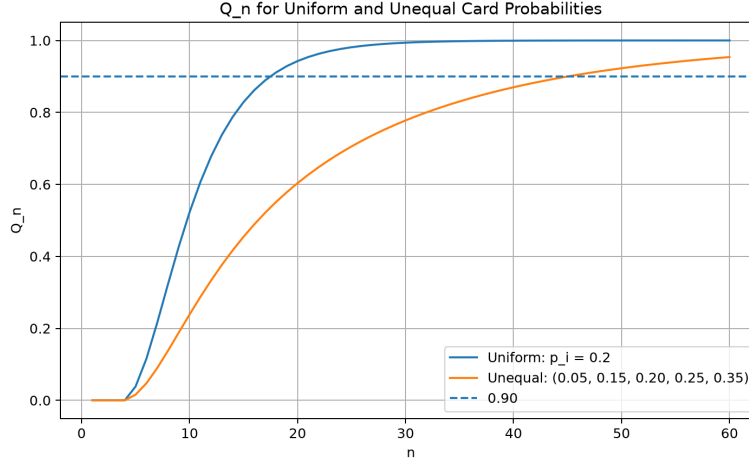


Figure 2: Q_n for uniform and unequal card probabilities.

From the graph,

$$Q_n^{\text{uniform}} > Q_n^{\text{unequal}}$$

for the relevant values of n .

In the unequal case, the rarest card has probability

$$p_1 = 0.05,$$

so it is much more difficult to collect all five card types. In the uniform case, every type has probability

$$p_i = 0.20,$$

so no single card type becomes a bottleneck. Therefore, the probability of collecting all five types increases much faster in the uniform case.